

Difference between Python and Java

Parameters	Python	Java
Code	Python has generally fewer lines of code.	Java has long lines of code.
Framework	Compare to JAVA, Python has a lower number of Frameworks. Popular ones are Django and Flask.	Java has a large number of Frameworks. Popular ones are Spring, Hibernate, etc.
Syntax	The syntax is easy to remember almost similar to human language.	The syntax is complex as it throws errors if you miss semicolons or curly braces.
Key Features	Less line no of code, Rapid deployment, and dynamic typing.	Self-memory management, Robust, Platform independent
Speed	Python is slower since it uses an interpreter and also determines the data type at run time.	Java is faster in speed as compared to python.
Databases	Python's database access layers are weaker than Java's JDBC. This is why it is rarely used in enterprises.	(JDBC)Java Database Connectivity is the most popular and widely used to connect with databases.
Machine Learning Libraries	Tensorflow, Pytorch.	Weka, Mallet, Deeplearning4j, MOA
Multiple Inheritance	Python supports multiple Inheritance.	Java partially supports Multiple Inheritance through interfaces.

Differences with Code Examples

	Python	Java
Program Setup & Printing	<pre>print("Hello!")</pre> Pure action. Just 1 line	<pre>public class Main { public static void main(String[] args) { System.out.println("Hello, World!"); } }</pre> High setup. Requires a Class and a Method container just to start.
Variables and Typing	<pre>x = 10 print(x) x = "Hello" print(x)</pre> Python figures out the type automatically and lets you change it.	<pre>public class Main { public static void main(String[] args) { int x = 10; System.out.println(x); // Note: You cannot reassign x to a // String like in Python // x = "Hello"; would cause a // compilation error } }</pre> Java forces you to declare the type ("int"). It can never hold text later.
If/Else Statements (Decision Making)	<pre>score = 85 if score > 50: print("Passed") print("Great job")</pre> Uses an if statement followed by code lines that are indented.	<pre>public class Main { public static void main(String[] args) { int score = 85; if (score > 50) { System.out.println("Passed"); System.out.println("Great job"); } } }</pre> Uses parentheses () for the condition, curly braces {} to wrap the actions, and semicolons ; at the end of lines.
For Loops (Counting to 3)	<pre>for i in range(1, 4): print(i)</pre>	<pre>public class Main { public static void main(String[] args) {</pre>

	Python prioritizes human readability by using simple, natural English phrases.	<pre> for (int i = 1; i <= 3; i++) { System.out.println(i); } } } </pre> Java uses a traditional, explicit computer-science formula that tracks the starting value, stopping condition, and step increment manually
Lists vs. Arrays (Storing Collections)	# A Python list can hold integers, strings, and floats together <pre>my_list = [1, "Apple", 3.5]</pre> # You can easily add items whenever you want <pre>my_list.append("Banana") print(my_list)</pre> Python lists are dynamic, flexible objects by default.	<pre> import java.util.ArrayList; public class Main { public static void main(String[] args) { // Standard arrays are fixed in size. // To get flexibility like Python, // Java uses an ArrayList, but it must still specify a single data type (String). ArrayList<String> myList = new ArrayList<>(); myList.add("Apple"); myList.add("Banana"); System.out.println(myList); } } </pre> Standard Java arrays are highly rigid, locked to a single data type, and fixed in size to make memory management faster and safer.
Writing Functions / Methods	# Python uses 'def' and doesn't care what type of data is returned <pre> def add_numbers(a, b): return a + b result = add_numbers(5,10) print(result) </pre>	<pre> public class Main { // Java requires you to state the type for inputs (int a, int b) // as well as the output type (int) before the method name public static int addNumbers(int a, int b) { return a + b; } public static void main(String[] args) { int result = addNumbers(5, 10); System.out.println(result); } } </pre>